

Embedded Systems and IoT

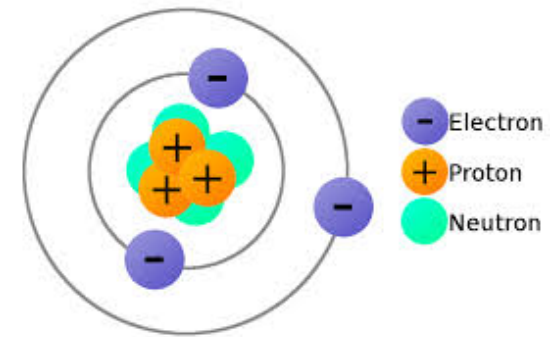
Ingegneria e Scienze Informatiche - UNIBO
a.a 2024/2025
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[Lab Notes]

BASIC ELECTRONICS

ELECTRIC CHARGE

- Elementary model of the atom:
 - **Electrons** – particles with negative charge
 - **Protons** – particles with positive charge
 - **Neutrons** – no charge
- The electron's **charge** equals that of the proton (opposite sign) and is denoted by **Q**, measured in Coulombs.
 - by definition, 1 C is the electric charge of 6.25×10^{18} electrons.
- Atoms normally have equal numbers of electrons and protons.
 - some electrons in the outer level can become *free electrons* by gaining energy from the environment



CONDUCTORS, SEMICONDUCTORS, INSULATORS

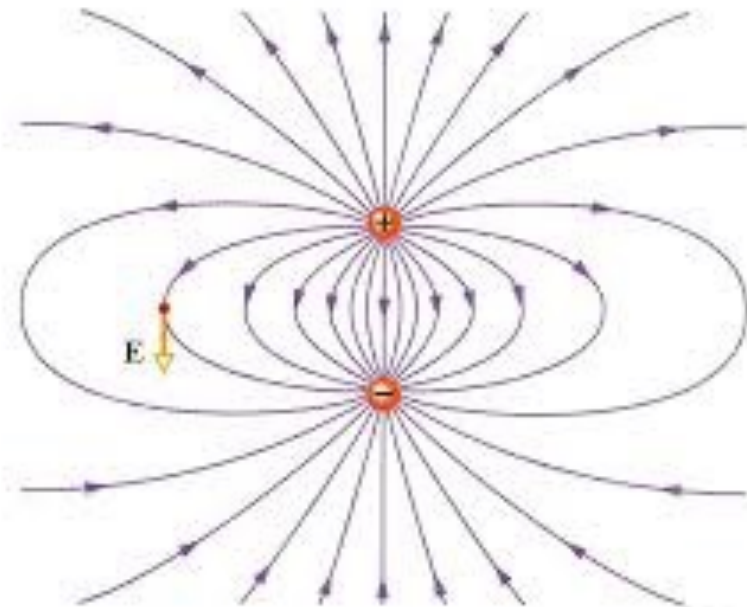
- Materials can be classified according to the quantity of free electrons
 - **Conductors** (e.g., silver, copper)
 - many free electrons.
 - **Semiconductors** (e.g., silicon)
 - have fewer.
 - **Insulators**
 - have very few.

STATIC ELECTRICITY

- Positive or negative *ions* form when neutral atoms gain or lose electrons
- A material with excess electrons has a *negative net charge*; excess protons give a *positive net charge*
- **Static electricity** is the presence of a net positive or negative charge

ELECTRIC FIELD

- Opposite charges attract, like charges repel
- An **electric field** **E** exerts forces **F** between charges



$$\mathbf{E} = \lim_{q \rightarrow 0} \frac{\mathbf{F}}{q}$$

VOLTAGE (POTENTIAL DIFFERENCE)

- To separate a positive and negative charge, *energy (work) is required*: opposite charges have a certain amount of *potential energy* that depends on their distance
- The difference in potential energy per unit charge is the **voltage (V)**:

$$V ::= W/Q$$

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{l}$$

where **W** is the energy (in Joules) and **Q** is the charge (in Coulombs).

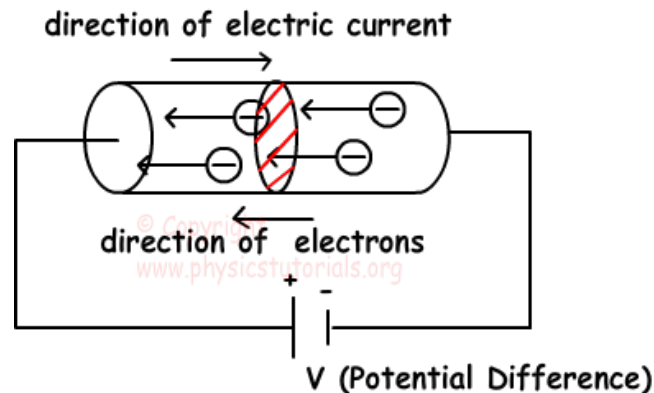
- *1 Volt is the potential difference (voltage) between two points when the energy of 1 joule is needed to move the charge of 1 Coulomb between the two points*

VOLTAGE SOURCES

- A **voltage generator** (electromotive force, EMF) provides potential energy, moving charges from lower to higher potential.
- Examples:
 - *Battery*: chemical to electrical energy
 - *Electronic power supply*: AC mains to DC
 - *Solar cell*: light to electrical energy (photovoltaic)
 - *Electric generator*: mechanical to electrical energy (electromagnetic induction)

ELECTRIC CURRENT (I)

- Free electrons in a material move randomly.
- Applying a voltage causes a net movement of electrons from negative to positive terminal.
- **Electric current (I)** is the ordered flow of electric charge



- Measured as the quantity of electric charge flowing over time, in **Ampères (A)**
 - $1 \text{ A} = 1 \text{ C of charge passing a point in } 1 \text{ s.}$

$$I ::= Q/t$$

$$I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

DIRECT AND ALTERNATING CURRENT

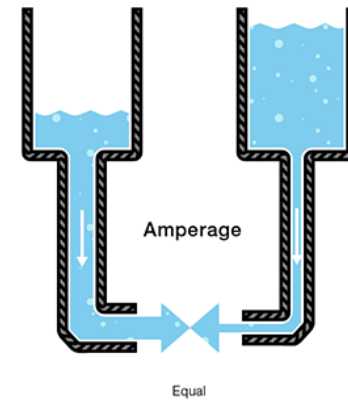
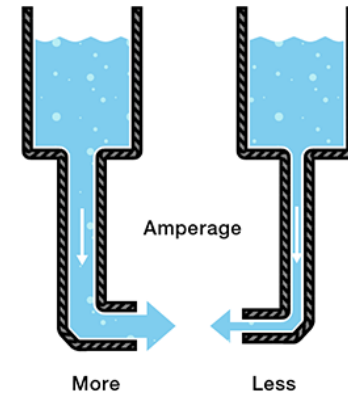
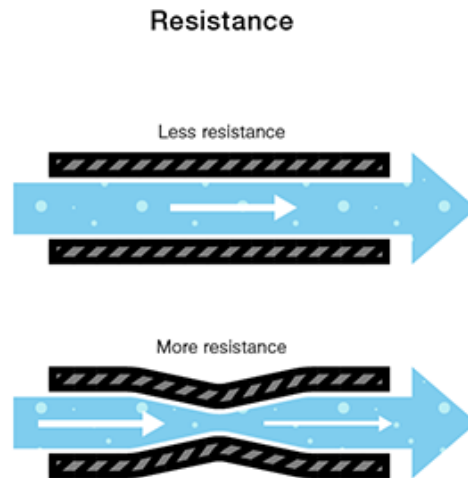
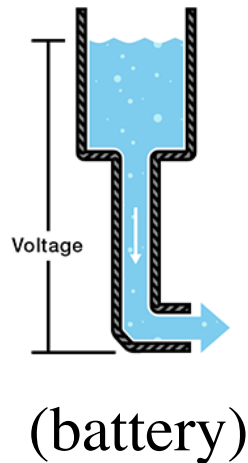
- **Direct Current (DC)**
 - current flows in one direction (e.g., batteries).
- **Alternating Current (AC)**
 - current changes direction periodically.
 - $I = I_0 \sin(2\pi f \cdot t)$, where f is the frequency
 - $f = 50 \text{ Hz}$ (Europe), 60 Hz (USA)

RESISTANCE

- Moving electrons collide with atoms, losing energy.
- **Resistance (R):** opposition to current flow, measured in **Ohms (Ω)**
 - 1 Ω when 1 A flows with 1 V applied.
- Conductance (G) is the inverse: $G = 1/R$
 - measured in Siemens (S)

HYDRAULIC ANALOGY

- Charge $\approx$ Water
- Voltage $\approx$ Pressure
- Current $\approx$ Flow



POWER

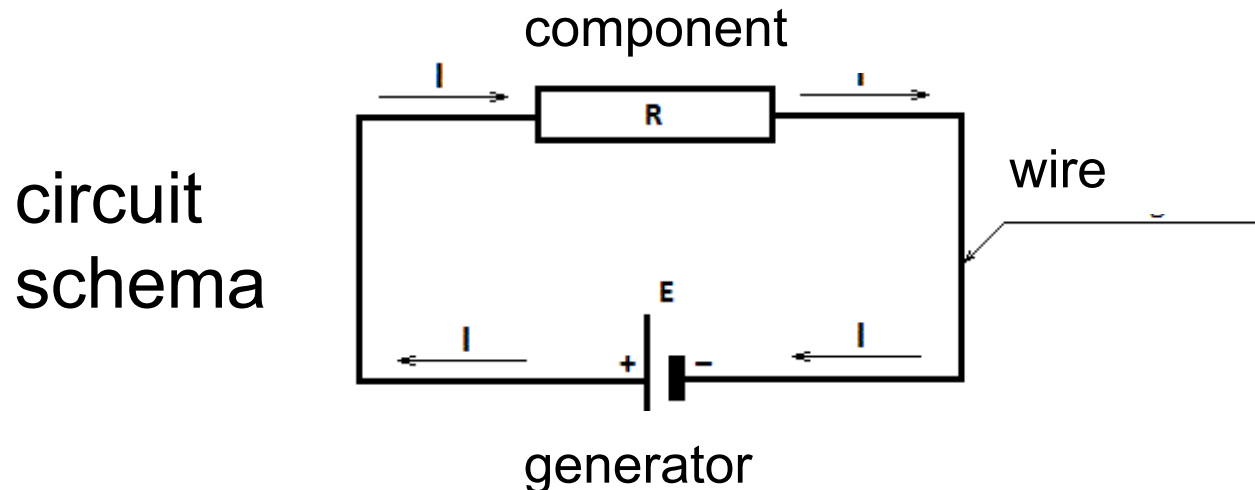
- Power (P) is the *rate of energy use*:

$$P ::= W / t$$

- Measured in **Watt**, that is Joules/second
 - 1 Watt is the quantity of power when 1 Joule is consumed in 1 second
- Energy consumed can be calculated given the power P and the time t => $W = P \times t$
 - energy in Joule also measured in wattsecond (Ws), watthour (Wh), kilowatthour (kWh)
 - e.g., a 100 W lamp on for 10 h consumes 1 kWh.

ELECTRIC CIRCUIT

- Composed of:
 - **Voltage source(s)**
 - **Electrical/electronic loads (components)**
 - **Connections allowing current to flow (wires)**

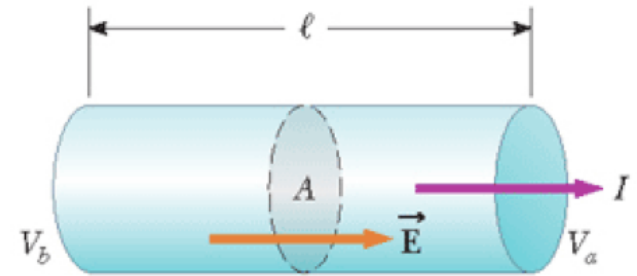


- The circuit must be **closed** for current to flow
- Components must be connected respecting their polarity, according to the current flow

OHM'S LAW

- Law relating V , I , R for many materials:

$$V = I \times R$$



- Resistance depends on material and geometry.
- Then:
 - in a circuit with some resistance R , by increasing/decreasing V , we linearly increase/decrease the current I
 - in a circuit with some fixed voltage V , by increasing/decreasing R , we linearly increase/decrease the current I

POWER IN A CIRCUIT (JOULE EFFECT)

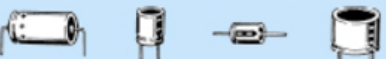
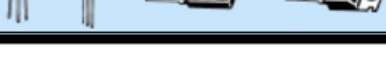
- Heat generated when current flows through a resistor, because of collisions inside atoms
- By using Ohm's law

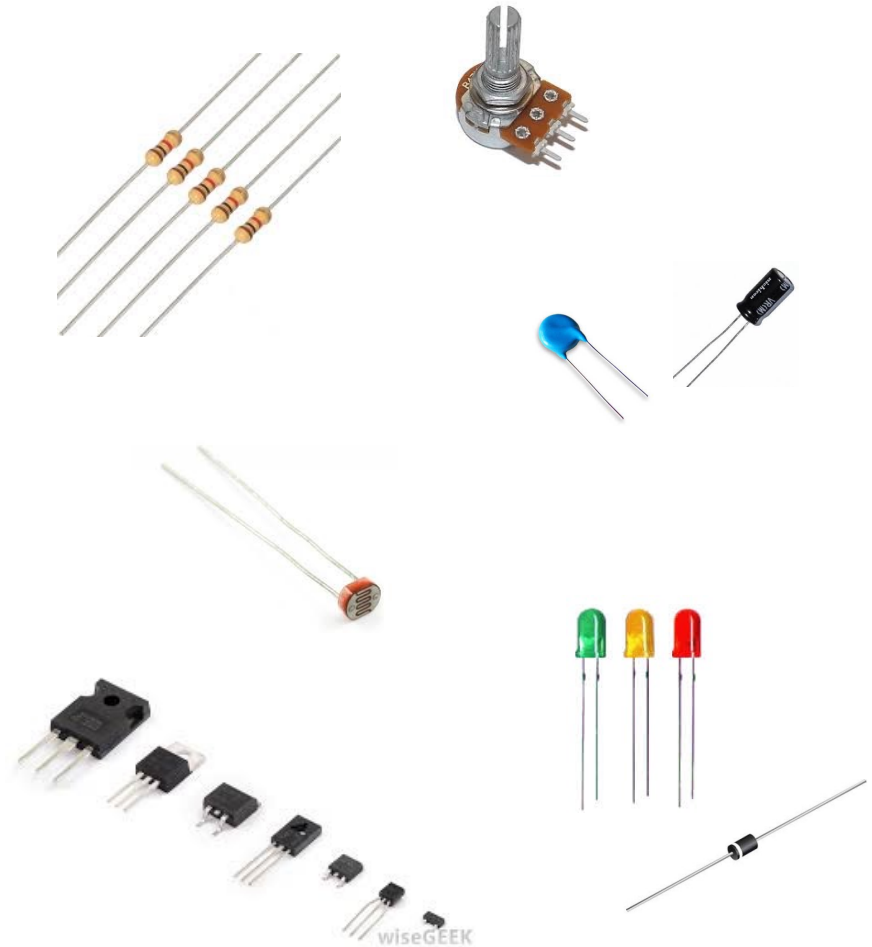
$$P = V \times I = I^2 \times R = V^2/R$$

ELECTRONIC COMPONENTS

- Components control current / voltage present in a circuit
 - Passive: do not supply energy
 - resistors, capacitors, inductors, diodes
 - also called impedances
 - Active: supply energy
 - transistors, integrated circuits.

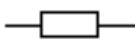
ELECTRONIC COMPONENTS

symbol	name	physical appearance
	R resistance	
	R trimmer	
	R potentiometer	
	FR photoresistor	
	C capacitor	
	C compensator	
	C elec. capacitor	
	DS diod	
	DZ zener diod	
	DV DIOODO VARICAP	
	DL led	
	FD photodiod	
	TR TRANSISTOR	
	FT FET	



RESISTORS

- **Resistors**

- Provide a voltage drop when current flows.
- Measured in Ohms (Ω).
- symbols  

- Types

- fixed, variable (potentiometers)

- Color code indicates resistance and tolerance.

- 4 band or 5 band — defining the resistance value
 - e.g. in 4 band
 - each color/band => a digit
 - first digit, second digit, multiplier (number of zeros)
 - fourth stripe: tolerance (5%, 10%, 20%)

RESISTORS COLOR CODE

Resistor Color Code - 4 Band, 5 Band and 6 Band

www.elcircuits.com

Color	1st Band	2nd Band	3rd Band	Multiplier	Tolerance	Temp. Coefficient
Black	0	0	0	X 1Ω		
Brown	1	1	1	X 10Ω	± 1%	100ppm
Red	2	2	2	X 100Ω	± 2%	50ppm
Orange	3	3	3	X 1KΩ		15ppm
Yellow	4	4	4	X 10KΩ		25ppm
Green	5	5	5	X 100KΩ	± 0.5%	
Blue	6	6	6	X 1MΩ	± 0.25%	
Violet	7	7	7	X 10MΩ	± 0.1%	
Grey	8	8	8		± 0.05%	
White	9	9	9			
Gold				X 0.1Ω	± 5%	
Silver				X 0.01Ω	± 10%	

www.elcircuits.com

4 BAND CASE

	1 st Digit	2 nd Digit	Multiplier	Tolerance
Black	0	0	x 1	
Brown	1	1	x10	±1%
Red	2	2	x10 ²	±2%
Orange	3	3	x10 ³	±3%
Yellow	4	4	x10 ⁴	±4%
Green	5	5	x10 ⁵	±0.5%
Blue	6	6	x10 ⁶	±0.25%
Violet	7	7	x10 ⁷	±0.1%
Grey	8	8	x10 ⁸	±0.05%
White	9	9	x10 ⁹	
Gold			x10 ⁻¹	±5%
Silver			x10 ⁻²	±10%

CAPACITORS AND INDUCTORS

- **Capacitors**

- Store electric charge and voltage
- Capacitance measured in Farads (F).
- Typical values: microfarads or less.
- symbol: 

- **Inductors**

- Generate a magnetic field when current flows.
- Used in transformers, relays, motors.
- Pass DC but oppose AC depending on frequency.
- symbol: 

DIODEs AND LEDs

- **Diod**

- semiconductor allowing current in one direction only.
- must observe polarity (anode +, cathode –).
- Used for rectification, detection, etc.
- symbol: 

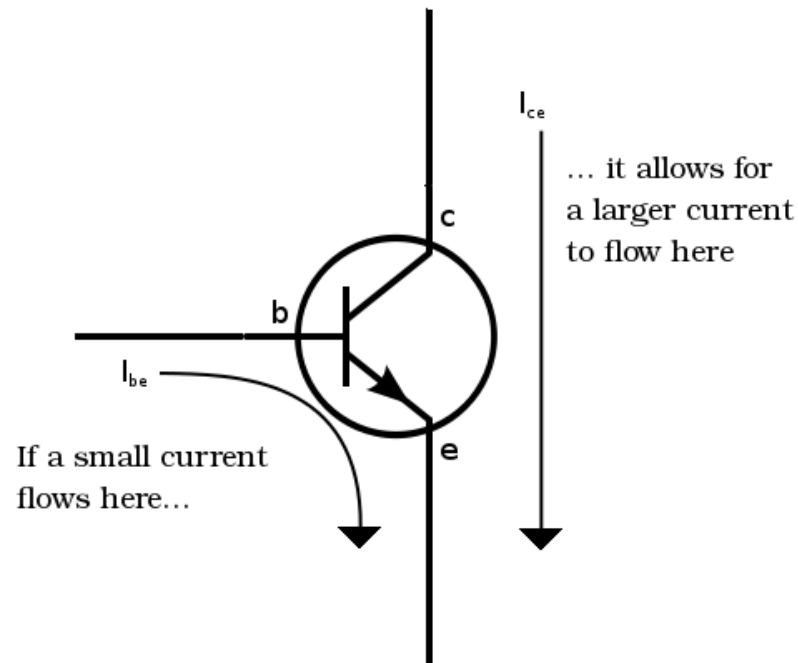
- **LED (Light Emitting Diode)**

- diode that emits light
- symbol: 

TRANSISTORS

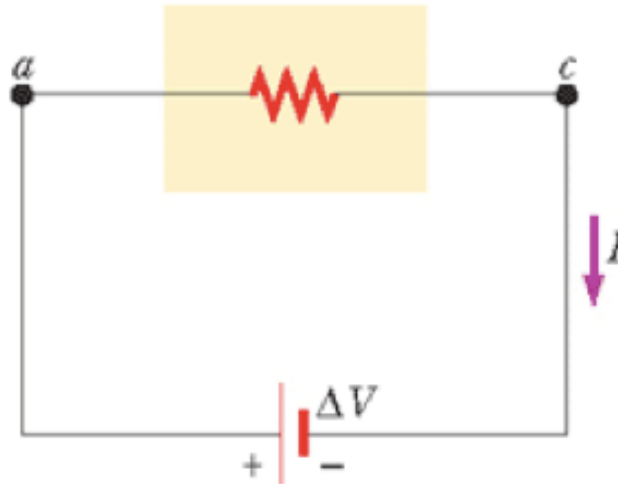
- **Transistors**

- Active component with three terminals: collector, base, emitter.
- Used for amplification, switching, driving power devices.
- symbols:   



CIRCUIT ANALYSIS

- Current flows from higher to lower potential (conventional current)



- For a resistor obeying Ohm's law:

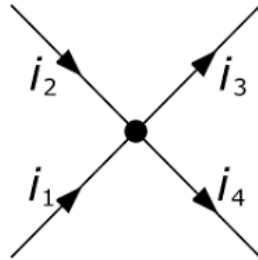
$$I = \Delta V / R$$

- Assuming negligible wire resistance and internal battery resistance.
 - ideal generator = no internal resistance

KIRCHHOFF'S LAWS

- **Current Law (Node Law)**

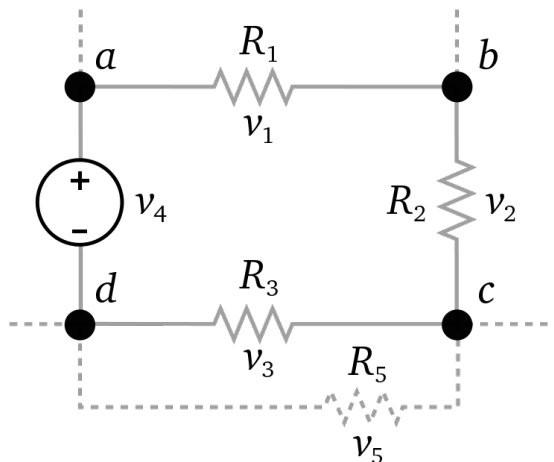
- the algebraic sum of currents entering a node is zero.



$$i_1 + i_2 = i_3 + i_4$$

- **Voltage Law (Loop Law)**

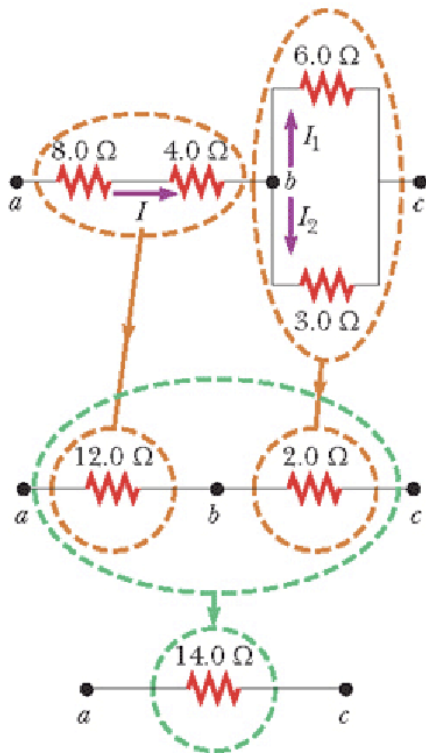
- the algebraic sum of voltages around a closed loop is zero.



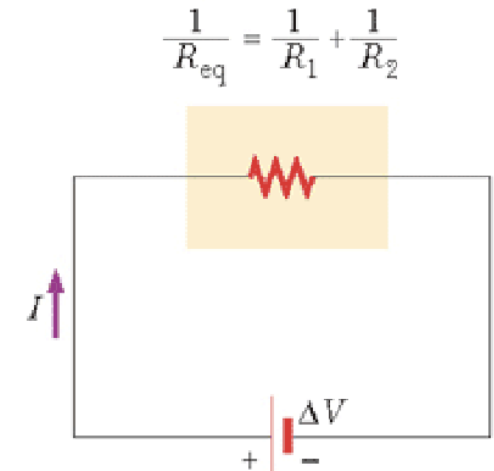
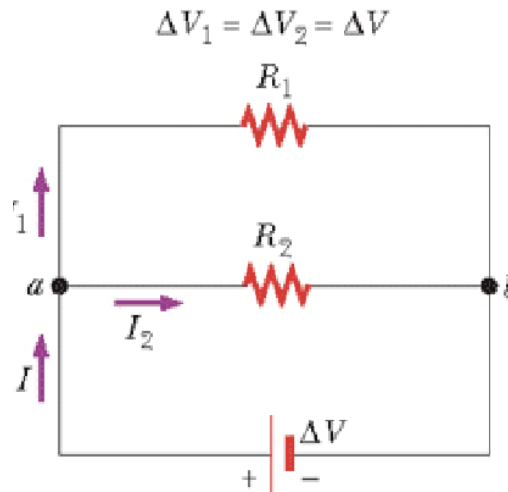
$$v_1 + v_2 + v_3 - v_4 = 0$$

RESISTORS IN SERIES AND PARALLEL

- Using Kirchhoff's laws
 - resistors in series: $R_{eq} = R_1 + R_2$
 - resistors in parallel: $R_{eq} = 1/(1/R_1 + 1/R_2) = R_1 * R_2 / (R_1 + R_2)$



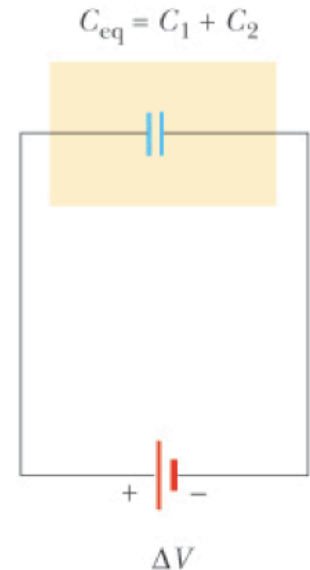
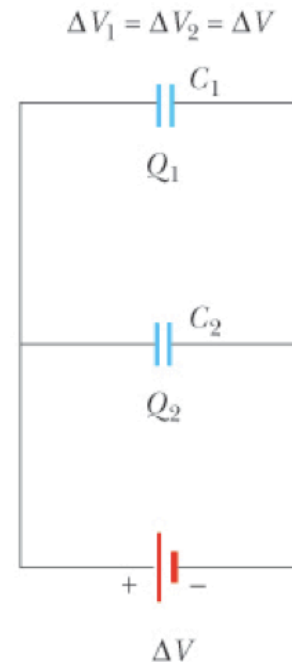
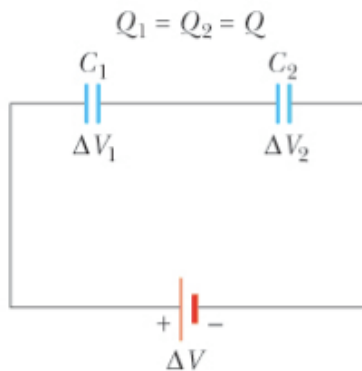
(in series)



(in parallel)

CAPACITORS IN SERIES AND PARALLEL

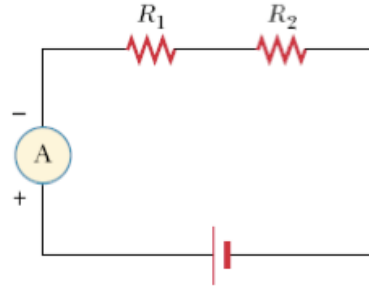
- For capacitors - using Kirchhoff's laws and knowing that $V = Q/C$:
 - capacitors in series: $C_{eq} = 1/(1/C_1 + 1/C_2) = C_1 * C_2 / (C_1 + C_2)$
 - capacitors in parallel: $C_{eq} = C_1 + C_2$



MEASURING INSTRUMENTS

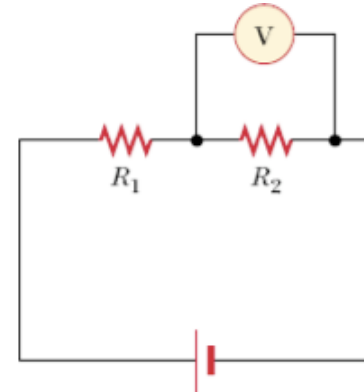
- **Ammeter**

- measures current
- connected in series
- very low internal resistance



- **Voltmeter**

- measures voltage
- connected in parallel
- very high internal resistance



- **Multimeter/Tester:** combines functions

